

TPC-400: Third-Family Replication of a Finite C1 Endpoint Microgrid

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Abstract

TPC-399 found a finite separation between same-law mean transfer and origin uniformity: the probes through $\lambda = 31/32$ were origin-stable, while the endpoint $\lambda = 1$ was not. TPC-400 tests the next response-blind question on a third coordinate-disjoint affine family. Its 96-row panel compares each current same-law cohort mean with a hash-locked all-origin mean from TPC-399. Every one of the 16 law/normalization cells passes both the three-percent calibration and holdout transfer caps, and all 16 within-family transfers pass. At the same time, $\lambda = 1$ fails the one-percent origin-spread rule in all four normalizations, with maximum spreads between 0.0536045 and 0.0538907. The endpoint calibration discrepancy approaches the declared cross-family boundary but remains below it. This is finite replication and obstruction evidence, not an arithmetic, asymptotic, Route-A/Route-B, or twin-prime theorem.

1 Question and claim boundary

TPC-399 supplied a frozen same-law interface from a second fresh family and showed that close cohort means can coexist with endpoint origin instability. The present paper asks whether that split survives on a third family, with all roles and probes fixed before the new response is read.

Every object below is finite. In particular, a fractional coefficient in a matrix interpolation is not asserted to be an arithmetic character, a prime sign law, or a source weight. The claim firewall is

$$\text{ARITHMETIC_ADVANCE=NO,} \quad \text{FIXED_POWER_CREDIT=0,} \quad \text{FULL_GATE_B=OPEN.}$$

The official Session evaluator files are absent from this checkout, so the local proof, independent checker, and Bridge-B artifacts provide finite consistency evidence only.

2 Finite construction

For a prime $p \in (Q, 2Q]$, set $Q = 8192$, $H = 66$, exponent one, and $\beta = 2$. The component used in the certificate is

$$K_p(u, v) = p(p/Q)^2 \frac{H^2}{H^2 + (u - v)^2} \left(\mathbf{1}_{p|u-v} - \frac{1}{p-1} \right) \\ \cdot \mathbf{1}_{u \neq v} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid v}.$$

Let

$$G(u) = \sum_{p \in (Q, 2Q]} \sum_v K_p(u, v)^2.$$

If the shell is indexed increasingly, define

$$M_+ = \sum_p K_p, \quad M_{\text{alt}} = \sum_p (-1)^{\text{index}(p)} K_p.$$

The four probes are the exact finite combinations

$$M_\lambda = (1 - \lambda)M_+ + \lambda M_{\text{alt}}, \quad \lambda \in \left\{ \frac{7}{8}, \frac{15}{16}, \frac{31}{32}, 1 \right\}.$$

Table 1: TPC-400 finite panel. Counts are across the four laws.

normalization	origin	cross-cal	cross-hold	within	max R
local diagonal	3/4	4/4	4/4	4/4	0.0536044969
pooled train scalar	3/4	4/4	4/4	4/4	0.0538721097
origin scalar	3/4	4/4	4/4	4/4	0.0538906727
frozen train-1024 scalar	3/4	4/4	4/4	4/4	0.0538721097

This is a finite linear-algebra identity, not an arithmetic interpolation theorem.

The response-blind third-family grid is

$$a_j = 7600001 + 401j, \quad 0 \leq j < 41,$$

with selected indices (0, 8, 16, 24, 32, 40) and origins

$$7600001, 7603209, 7606417, 7609625, 7612833, 7616041.$$

The first three are calibration origins and the last three are holdout origins. Every window has $N = 1024$ points, split into eight blocks of length 128; the fixed $c = 3$ band retains block pairs at distance at most three.

3 Diagnostics and frozen parent interface

Four normalizations are recorded: local diagonal, pooled calibration scalar, current-origin scalar, and first-calibration frozen scalar. If $S_\lambda(o)$ denotes the masked spectral diagnostic, its origin spread is

$$R_\lambda = \frac{\max_o S_\lambda(o) - \min_o S_\lambda(o)}{\text{mean}_o S_\lambda(o)}.$$

The origin cap is $R_\lambda \leq 0.01$. Finite spectral and Schur caps are 0.64 and 0.83.

TPC-399 is imported only through its hash-locked code and canonical certificate. For each law and normalization, let $B_\lambda^{(399)}$ be its recorded all-origin mean. The current cohort errors are

$$E_{\text{cal}} = \frac{\text{mean}_{o \in \mathcal{C}} S_\lambda(o)}{B_\lambda^{(399)}} - 1, \quad E_{\text{hold}} = \frac{\text{mean}_{o \in \mathcal{H}} S_\lambda(o)}{B_\lambda^{(399)}} - 1.$$

Both use cap 0.03. The within-family error is

$$E_{\text{within}} = \frac{\text{mean}_{o \in \mathcal{H}} S_\lambda(o)}{\text{mean}_{o \in \mathcal{C}} S_\lambda(o)} - 1,$$

also tested at cap 0.03. No segment or threshold fit is performed.

The parent locks are

```
code: 6b65f30fd6aa3f54e58596635a1248c892c01eb71d4156a37578bb71a1079d2b
certificate: 6f632add733947838c4268969748068633b2b85fadbd8fba7c21a146d98b7896
```

The producer accumulates the shell in ascending order. An independent checker accumulates it in descending order and recomputes all aggregates. At the exact anchor [7600001, 7600014) with $Q = 8$ and shell {11, 13}, exact ‘Fraction’ arithmetic verifies positive geometry, symmetry, and all four interpolation identities.

4 Results

The interior laws 7/8, 15/16, and 31/32 pass the one-percent origin-spread cap under all four normalizations. The endpoint law 1 passes none, giving 12 of 16 origin-stable cells. The maximum absolute cross-family calibration errors, in local, pooled, origin, and frozen order, are

$$0.0242418805, \quad 0.0277738760, \quad 0.0277699591, \quad 0.0277815666,$$

and the corresponding holdout maxima are

0.0001317872, 0.0024016863, 0.0023850574, 0.0024091870.

All 16 cells pass both cross-family tests. The maximum absolute within-family transfer errors in the same order are 0.0236215933, 0.0246865486, 0.0246990112, and 0.0246865486; all 16 pass. No spectral or Schur row fails among the 96 rows. The spectral and Schur values are reproducible float64 observations; no interval eigenvalue enclosure or propagated rounding certificate is claimed.

5 Interpretation and route ledger

The strongest positive result is a response-blind third-family replication: the frozen TPC-399 same-law means transfer to the current family in every law, normalization, and cohort role at the declared three-percent scale. The strongest obstruction is that the endpoint remains origin-unstable, with a 5.36–5.39 percent spread. The endpoint calibration error is close to the cross-family cap, so the finite panel does not justify a stronger uniformity claim.

The reusable structure is a direct hash-locked same-law interface, exact finite matrix interpolation, coordinate-disjoint calibration/holdout families with explicit prior-interval checks, reverse-order replay, and separate origin and transfer gates. The next response-blind clue is

ROUND2_CLUE=TEST_C1_ENDPOINT_MICROGRID_FOURTH_FAMILY_REPLICATION.

The theorem ledger is intentionally conservative:

status	finite scope
PROVED_EXACT_FINITE	selection, disjointness, hashes, anchor identities
NUMERICAL_OBSERVATION	96 float64 rows and stated aggregate observations
OPEN	source-valid uniformity, growing bounds, arithmetic L^2
NONE	arithmetic advance, fixed-power credit, twin-prime result

Route-A is not officially evaluated because its Session evaluator is absent; Route-B remains open. The certificate makes no claim about a source-uniform estimate, an arithmetic L^2 bound, a growing operator, or the twin-prime conjecture.

Reproduction

The project contains the required README, code, experiments, results, notes, proof package, and Bridge-B artifacts. The canonical certificate is `results/tpc400_certificate.json`; the release contains byte-identical `paper/main.pdf` and `paper/paper.pdf`. The README lists normal and optimized commands for all checks.